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# Step 1: Import required libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Step 2: Upload an image
uploaded = files.upload()

# Step 3: Read the uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)

# Step 4: Convert image from BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Step 5: Create a blurred version of the image
blurred = cv2.GaussianBlur(image, (5, 5), 0)

# Step 6: Perform Unsharp Masking
# Original image + detail mask
sharpened = cv2.addWeighted(image, 1.5, blurred, -0.5, 0)

# Step 7: Convert sharpened image to RGB
sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)

# Step 8: Display original, blurred and sharpened images
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(cv2.cvtColor(blurred, cv2.COLOR_BGR2RGB))
plt.title("Blurred Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sharpened_rgb)
plt.title("Unsharp Masking")
plt.axis("off")

plt.show()
```

every-detail...p-photo.jpg

**every-detail-of-a-sleek-modern-car-captured-in-close-up-photo.jpg**(image/jpeg) - 35579 bytes, last modified: 9/1/2026 - 100% done

Saving every-detail-of-a-sleek-modern-car-captured-in-close-up-photo.jpg to every-detail-of-a-sleek-modern-car-captured-in-close-up-photo (8).jpg

Original Image



Blurred Image



Unsharp Masking

